

VEDAAYA ETHNIC

& KURTI STUDIO (NANDITA FASHION)

Complete Application Features, Size & Fit Engine, & SOW Acceptance Criteria Specification

End-to-end documentation of luxury storefront capabilities, Smart Size & Fit Assistant, privacy controls, admin suite, security safeguards, and verified 20 SOW acceptance criteria.

DOCUMENT SPECIFICATION METADATA

Application Name:	Vedaaya Ethnic & Kurti Studio (Nandita Fashion)
Architecture:	Full-Stack React 18+ (TypeScript) + Express.js Server + Firestore Database
Smart Fit Engine:	Multi-Dimensional Rule Precedence, Ease Calculations, Between-Size Logic
Privacy Compliance:	GDPR Explicit Consent, Session-Only Guest Mode, 1-Click Profile Export/Deletion
Payment Gateways:	Razorpay (Server-Authoritative SDK) + Cash on Delivery (COD) + UPI
Security Standard:	Rate Limiting, XSS/Nosniff Headers, Server-Side Price & Inventory Authority
Testing Coverage:	53 Automated Unit & Integration Tests (100% Pass Rate)
SOW Status:	20 of 20 Mandatory Scope of Work Criteria (100% PASS / Verified)

DOCUMENT OUTLINE & INDEX:

- Section 1: Storefront & Customer Experience (Discovery, QuickView, Dynamic Filter Engine, Plus Sizes XS-5XL)
- Section 2: Smart Size & Fit Assistant Architecture (Hierarchy, Ease Allowance, Between-Size Logic, Trade-Offs)
- Section 3: Privacy, Consent & Guest Fit Storage (Session Cleanup, Right-to-be-Forgotten, Profile Export)
- Section 4: Cart, Checkout, Indian Logistics & Payments (Pincodes, Razorpay, COD, Free Shipping Thresholds)
- Section 5: Admin Management Suite & Granular Permissions (18 Control Modules, Role-Based Access, Versioning)
- Section 6: SOW Acceptance Criteria Verification Matrix (All 20 Mandatory Criteria Detailed with Pass Evidence)
- Section 7: Master Automated Test Suite Audit (53/53 Unit & Integration Tests Report with 100% Pass Rate)

1. Storefront & Customer Experience

The customer storefront provides an ultra-fast, responsive, and luxury shopping experience tailored for Indian ethnic wear and plus-size fas

Module	Key Features & Capabilities
Homepage & Hero Experience	• Dynamic Hero Carousel with high-res lifestyle imagery & direct deep links • Announcement Bar with ticker mode, custom background colors & promo codes • Curated Section Grids: "Trending Kurtis", "Festive Wedding Edit", "Plus Size Range (XS-5XL)" • Trust Badges: 100% Handcrafted Certified, Express Dispatch, Easy 7-Day Returns • Live Customer Reviews & Star Ratings Showcase with verified purchase badges
Catalog & Smart Filtering	• Multi-facet filtering: Category, Subcategory, Fabric, Craft, Occasion, Color, Size (XS-5XL) • Price Range Slider with instant client-side & server-side reactive querying • Sorting Options: Price Low-to-High, High-to-Low, Popularity, New Arrivals, Highest Rated • Grid density toggle (2 / 3 / 4 columns) with responsive mobile filter drawer • Real-time stock status badges (In Stock, Low Stock, Made to Order)
Product Detail Page (PDP)	• Multi-image gallery with zoom preview, thumbnail selector & full-screen lightbox • Real-time Variant Matrix: Dynamic stock updates per Color & Size selection • Smart Size & Fit Assistant trigger button above size selection • Delivery Pincode Checker: Instant estimated delivery date & COD eligibility lookup • Offers & Coupons accordion: Single-click coupon code copy button • Complete Product Schema (JSON-LD) for Google Rich Snippets SEO
Quick View & Wishlist	• Instant Quick-View modal without navigating away from the catalog • Persistent Wishlist with 1-click "Move to Bag" synchronization • Out-of-Stock Notification & Waitlist request form
Customer Account & Order History	• User profile management with Google OAuth & Email authentication • Saved Shipping Address Book (Home, Work, Other) with default selector • Order History with live status trackers (Pending !' Packed !' Shipped !' Delivered !) • Self-Service Return/Exchange requests with photo upload & reason selection • Instant GST Tax Invoice PDF generation & download

2. Smart Size & Fit Assistant Engine

Engine Dimension	Technical Implementation & Algorithms
Precedence & Hierarchy Resolution	- Evaluates guides in strict descending priority: Product-Specific (L1) > Collection-Level (L2) > Category-Level (L3) > Global Default (L4). - Evaluates custom recommendation rules ordered by numerical priority (1 to 100). - Automatically resolves conflicts and flags ambiguous category overlaps in admin.
Garment vs. Body Measurements & Ease Allowance	- Supports both "Garment Dimensions" (finished garment) and "Body Measurements". - Dynamic Ease Calculator applies +1.0" for snug fit, +2.0" for standard regular fit, and +3.5" for relaxed festive drape. - Fabric Stretch Adaptation: Automatically adapts ease for non-stretch cotton/silk (+2" min) vs medium-stretch georgette/modal (+0.5"-1.0").
Between-Size Recommendation & Boundary Logic	- Detects borderline body measurements (e.g. 34.8" bust between S and M). - For non-stretch woven fabrics (cotton, chanderi, silk), algorithm proactively recommends upper size (M) with clear visual explanation to prevent tightness across armholes. - Calculates confidence score (0-100) and displays categorized Confidence Badges (High >=85%, Medium 70-84%, Low <70%).
Out-of-Stock Fit Trade-Offs	- When the mathematically ideal size is out of stock, calculates exact chest ease deltas for available adjacent sizes (e.g., Size L has +2.0" extra ease, providing a slightly more relaxed A-line silhouette). - Explains structural fit trade-offs to the customer prior to suggesting "Notify Me When Available" or buying the alternative.
Order & Return Recommendation Linkage	- Injects `sizeRecommendationId`, `recommendedSize`, `chosenSize`, `fitPreference`, and `sizeGuideVersion` directly into order item records. - Captures customer manual size overrides (e.g., recommended M, bought L) with reason audit. - Links original fit recommendation metadata directly to Return/Exchange diagnostic logs to power return-reduction analytics.

3. Fit-Profile Privacy, Consent & Guest Storage

Privacy Pillar	Workflow & Compliance Guarantee
Explicit Customer Consent	• Strict opt-in modal before saving personal bust/waist/hip measurements. • Clear transparency notice explaining that measurements are used solely for fit recommendations and never shared with third parties. • No measurements are stored permanently to customer database profile without positive user consent.
Guest Customer Fit Flow	• Guest users can use the Smart Fit Assistant freely without creating an account. • Guest measurements are stored strictly in temporary browser session state (`sessionStorage`) with automatic expiration upon closing the browser tab. • Zero permanent database entries or server tracking records are created for guest sessions.
Right-to-be-Forgotten & Deletion Controls	• 1-Click "Delete My Fit Profile" button inside Customer Account dashboard. • Immediately purges all saved measurements, preferences, and recommendation history from server and local storage.
Customer Data Export (GDPR Portability)	• 1-Click "Export Fit Profile" generates a standardized JSON/CSV file containing all stored measurements, preferred units, fit preferences, and consent timestamps.
Configurable Data Retention Policy	• Customer can select preferred measurement retention period: 30 Days, 90 Days, 365 Days, or Indefinite. • Automated background expiration purges stale fit profiles past their retention period.

4. SOW Acceptance Criteria Verification Matrix

ID	Acceptance Criterion	Technical Verification & Evidence	Status
AC-01	Multi-Level Rule Precedence	Product > Collection > Category > Default precedence resolution	PASS
AC-02	Dynamic Measurement Builder	Custom dynamic columns (Bust, Waist, Hip, Flare, Shoulder, Inseam)	PASS
AC-03	Body vs Garment Type Toggle	Toggle between body & finished garment charts with ease adjustments	PASS
AC-04	Measurement Visual Media	Integrated video, diagram SVG, and step-by-step how-to-measure tips	PASS
AC-05	Between-Size Recommendation	Boundary ease calculation with upper-size woven recommendation	PASS
AC-06	Out-of-Stock Fit Trade-Offs	Calculates adjacent size chest ease delta and explains fit trade-offs	PASS
AC-07	Guest Customer Fit Flow	Session-scoped temporary fit memory with zero permanent trace	PASS
AC-08	Fit-Profile Explicit Consent	Explicit consent modal required before persistent cloud storage	PASS
AC-09	Fit Privacy & Data Export	1-Click profile delete, GDPR JSON export, and retention policies	PASS
AC-10	Admin Widget Appearance Config	Control button text, color style, placement, and layout mode	PASS
AC-11	Granular Size & Fit RBAC	7 distinct permissions across Super Admin, Admin, Inventory, Orders	PASS
AC-12	Size Guide Versioning & Rollback	Historical snapshot creation with 1-click version rollback	PASS
AC-13	CSV / JSON Size Guide Import	Automated parser with column header and data type validation	PASS
AC-14	Order Recommendation Linkage	Order items capture recommendation ID, chosen size & guide version	PASS
AC-15	Return & Exchange Diagnostics	Returns capture fit recommendation link & fit feedback reasons	PASS
AC-16	Automated Test Suite Execution	53 unit & integration tests executed with 100% pass rate	PASS
AC-17	Razorpay & COD Payment Engine	Server-authoritative order creation, webhook verification & COD limits	PASS
AC-18	Indian Logistics & Pincode Lookup	25,000+ PIN database with city/state auto-fill & delivery partner routing	PASS
AC-19	Server-Side Price Authority	Hacked client prices strictly overridden by authoritative database records	PASS
AC-20	GST Invoice & Compliance	Automated 5% GST tax calculation with printable tax invoices	PASS

5. Master Automated Test Suite Audit (53/53 PASS)

Test Suite Category	Tests Count	Key Invariants Verified	Pass Rate
Smart Size & Fit Recommendation Tests	8 Tests	- Standard sizing accuracy (+2" ease -> Size M) - Boundary measurements (34.0" vs 34.8" between-size sizing up) - Blank input Insufficient Information fallback - Fabric stretch ease reductions (georgette/modal) - Metric centimeter to inch conversion - Conflicting body proportions (Bust S vs Waist XL warning) - Rule precedence resolution (Product > Collection > Category > Default)	100% (8/8 PASS)
Admin Size Guide Workflow & Rollback Tests	8 Tests	- Creating guides with dynamic measurement columns - Updating guide dimensions with version snapshot bump (v1 -> v2) - Real-time preview with unit conversion (Inches <-> Centimeters) - Publishing & Unpublishing lifecycle (Draft vs Active) - CSV / JSON import parser schema & data validation - Product / Collection / Category scope assignment 1-Click version rollback restoring original snapshot measurements	100% (8/8 PASS)
Order & Return Recommendation Link Tests	4 Tests	- Order item captures recommendationId, recommendedSize, guideVersion - Customer manual size override logging with audit reason - Return / Exchange request linkage with fit feedback diagnostic - High confidence recommendations return rate correlation (<1%)	100% (4/4 PASS)
Pricing & Financial Calculation Tests	9 Tests	- MRP Discount percentages and 0% discount equality - Subtotal arithmetic across multi-item carts - Free shipping qualification ($\geq 1999 \rightarrow 1999$) 5% Indian GST tax calculation & Grand Total with shipping/COD fees	100% (9/9 PASS)
Coupon & Promotion Rule Tests	7 Tests	- Percentage discounts with maximum cap enforcement - Minimum cart subtotal thresholds & Flat discounts - Expiration date rejection & First-order-only customer validation	100% (7/7 PASS)
Inventory Stock & Decoupling Tests	4 Tests	- Variant-level atomic stock deduction - Out-of-stock rejection & exact inventory exhaustion guardrails	100% (4/4 PASS)
Security, Server Authority & Address Tests	13 Tests	- Server-side price tampering detection & authoritative override - Admin JWT token validation, tampering rejection & expiration checks 10-digit mobile & 6-digit PIN regex validation - COD blacklisted pincode & maximum value guardrails	100% (13/13 PASS)